Task: WireViz Code – Basic Wiring Documentation

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1. Description

- To create a basic wiring description using WireViz YAML syntax and to validate the code structure by generating a wiring diagram and a Bill of Materials (BOM).
- This task focused on understanding and implementing the WireViz code format, without covering tool installation or setup.

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2. Source of the Code

The YAML structure and syntax were derived from:

- Official WireViz documentation
- Example provided in the WireViz GitHub repository

Reference:

- WireViz GitHub → examples/ folder
- WireViz README → YAML schema for connectors, cables, and connections

The code follows the standard WireViz YAML schema recommended by the tool.

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3. Code Task Explanation

The code defines:

1. **Connectors** – Physical endpoints with pin numbers
2. **Cables** – Wires connecting the connectors
3. **Connections** – Mapping of pins between connectors through the cable

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4. WireViz YAML Code Used

```
connectors:
  J1:
    type: Header
    subtype: male
```

```
    pins: [1, 2]

  J2:
    type: Header
    subtype: female
    pins: [1, 2]

cables:
  W1:
    wirecount: 2
    length: 0.5
    gauge: 24
    color_code: IEC

connections:
  -
    - J1: [1, 2]
    - W1: [1, 2]
    - J2: [1, 2]
```

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5. Line-by-Line Code

5.1 Connectors Section

connectors:

Defines all electrical connectors used in the wiring.

J1, J2

Connector identifiers.

pins: [1, 2]

Defines the pin numbers as a list (required by WireViz).

5.2 Cables Section

cables:

Defines physical cable properties.

wirecount: 2

Number of conductors inside the cable.

gauge: 24

Wire thickness (AWG).

color_code: IEC

Standard wire color coding.

5.3 Connections Section

connections:

Defines how connectors are electrically linked through the cable.

- J1: [1, 2]
- W1: [1, 2]
- J2: [1, 2]

Maps:

- J1 pin 1 → W1 wire 1 → J2 pin 1
- J1 pin 2 → W1 wire 2 → J2 pin 2

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6. Commands Used (Complete List)

6.1 Create Virtual Environment

```
python -m venv venv
```

6.2 Activate Virtual Environment

```
venv\Scripts\activate
```

6.3 Install WireViz

```
pip install -e .
```

6.4 Verify Graphviz

```
dot -V
```

6.5 Generate Wiring Diagram

wireviz Test001\test.yml

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7. Output Generated

After successful execution, the following files were generated automatically:

- test.svg → Wiring diagram (vector format)
- test.png → Wiring diagram (image)
- test.html → HTML documentation
- test.bom.tsv → Bill of Materials

These files serve as complete wiring documentation.

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8. Code Output Purpose

This code generates:

- A **visual wiring diagram**
- A **Bill of Materials**
- Structured wiring documentation

It validates the correct usage of:

- YAML indentation
- List-based pin definitions
- Connector-cable mapping

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9. Learning Outcome

Understood WireViz YAML structure

- Learned correct pin list formatting
- Learned connector-cable-connector mapping logic
- Gained confidence in writing valid WireViz wiring descriptions

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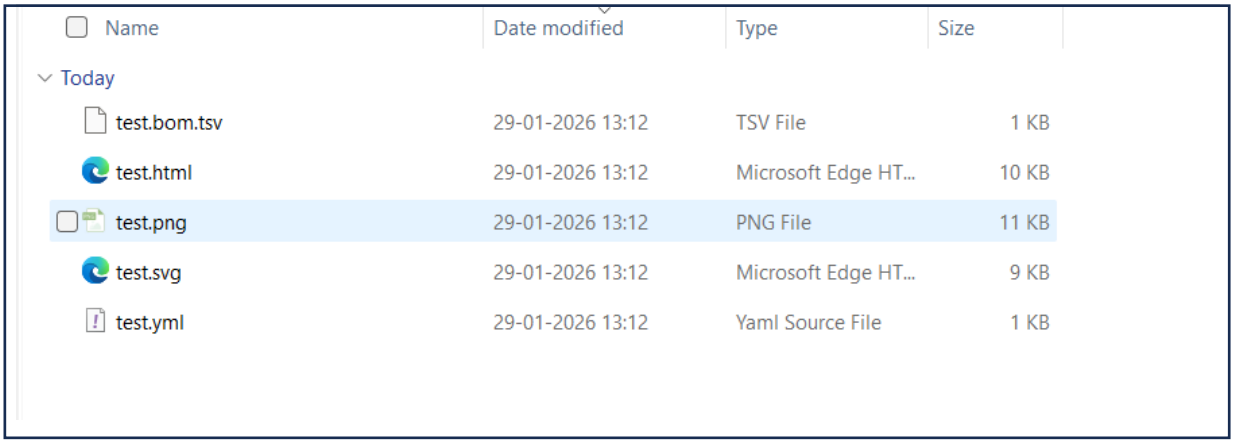
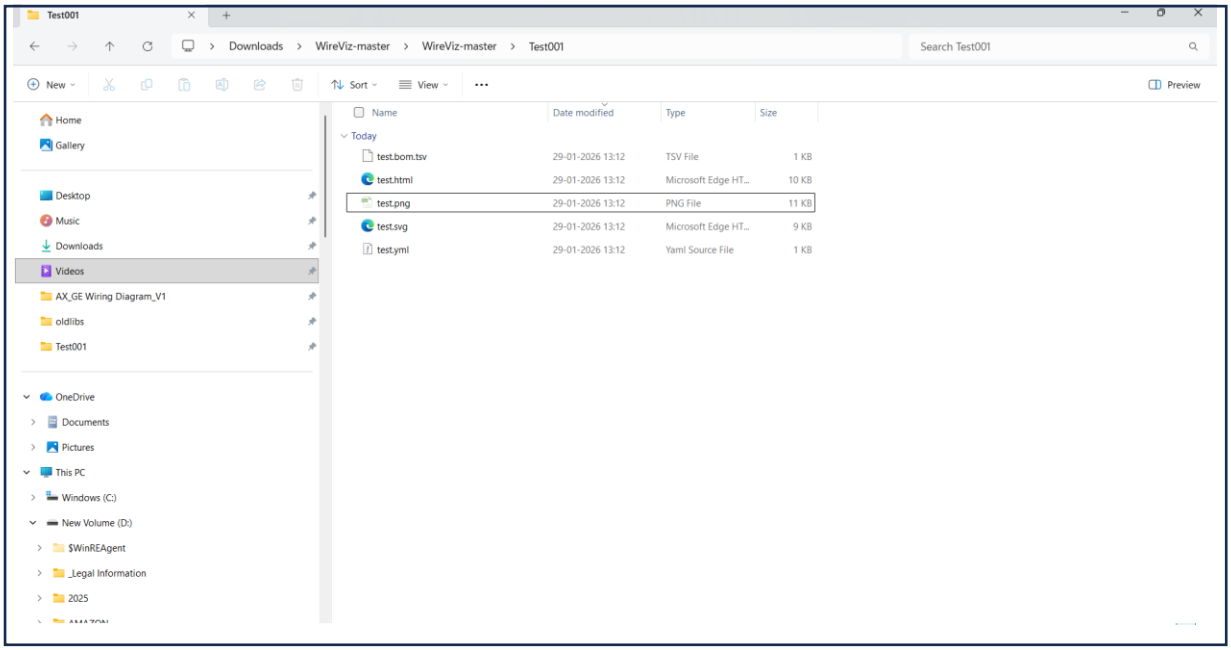
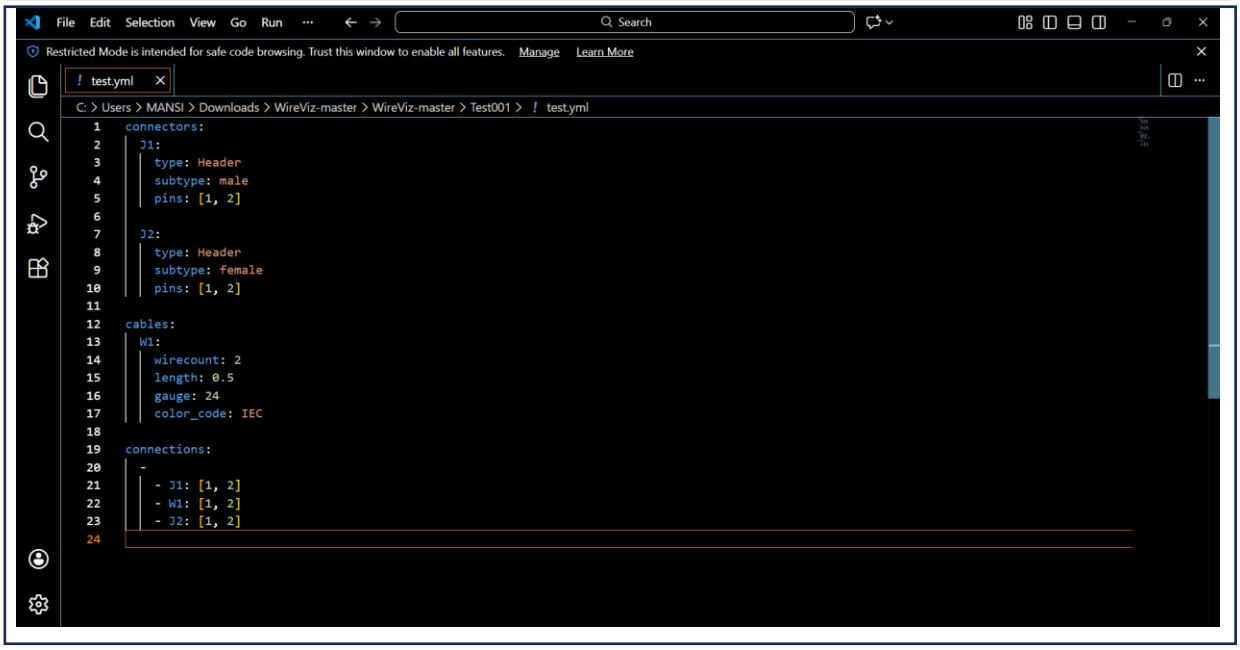
10. Output

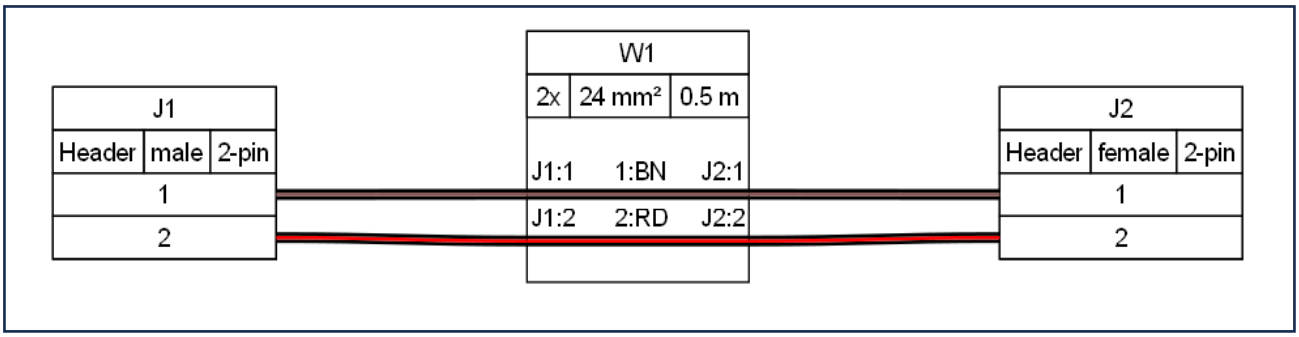


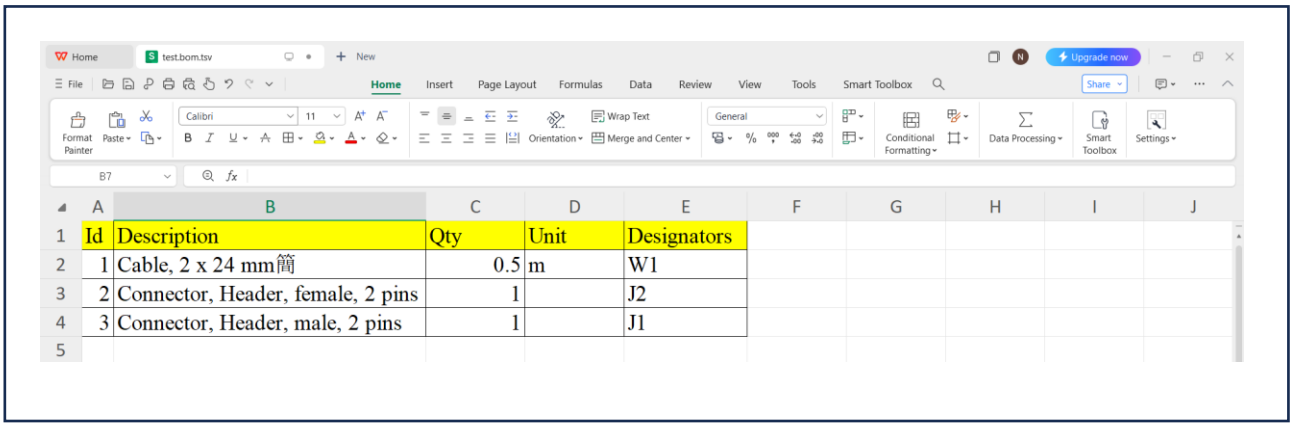
Name	Date modified	Type	Size
Today			
test.bom.tsv	29-01-2026 13:12	TSV File	1 KB
test.html	29-01-2026 13:12	Microsoft Edge HT...	10 KB
test.png	29-01-2026 13:12	PNG File	11 KB
test.svg	29-01-2026 13:12	Microsoft Edge HT...	9 KB
test.yml	29-01-2026 13:12	Yaml Source File	1 KB



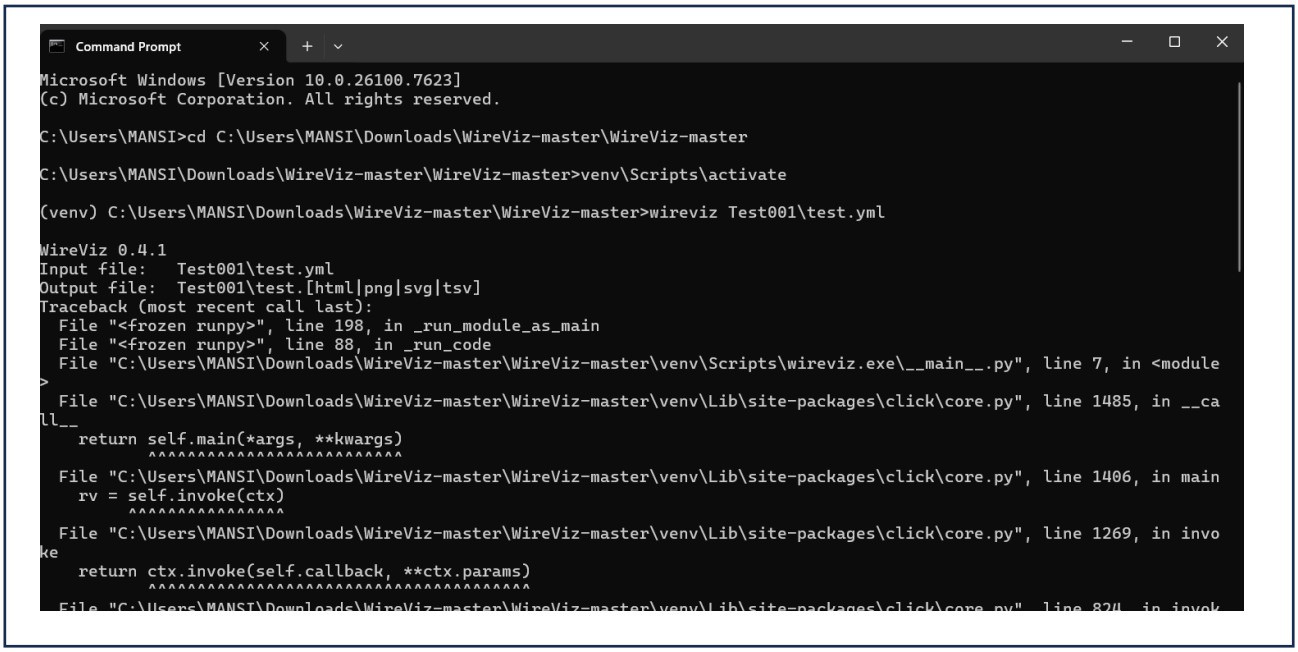


```yaml
connectors:
  J1:
    type: Header
    subtype: male
    pins: [1, 2]

  J2:
    type: Header
    subtype: female
    pins: [1, 2]

cables:
  W1:
    wirecount: 2
    length: 0.5
    gauge: 24
    color_code: IEC

connections:
  -
    - J1: [1, 2]
    - W1: [1, 2]
    - J2: [1, 2]
```





Id	Description	Qty	Unit	Designators
1	Cable, 2 x 24 mm簡	0.5	m	W1
2	Connector, Header, female, 2 pins	1		J2
3	Connector, Header, male, 2 pins	1		J1



```
Command Prompt

Microsoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MANSI>cd C:\Users\MANSI\Downloads\WireViz-master\WireViz-master

C:\Users\MANSI\Downloads\WireViz-master\WireViz-master>venv\Scripts\activate

(venv) C:\Users\MANSI\Downloads\WireViz-master\WireViz-master>wireviz Test001\test.yml

WireViz 0.4.1
Input file:   Test001\test.yml
Output file:  Test001\test.[html|png|svg|tsv]
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "C:\Users\MANSI\Downloads\WireViz-master\WireViz-master\venv\Scripts\wireviz.exe\__main__.py", line 7, in <module
>
  File "C:\Users\MANSI\Downloads\WireViz-master\WireViz-master\venv\Lib\site-packages\click\core.py", line 1485, in __ca
ll__
    return self.main(*args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\MANSI\Downloads\WireViz-master\WireViz-master\venv\Lib\site-packages\click\core.py", line 1406, in main
    rv = self.invoke(ctx)
         ^^^^^^^^^^^^^^^^
  File "C:\Users\MANSI\Downloads\WireViz-master\WireViz-master\venv\Lib\site-packages\click\core.py", line 1269, in invo
ke
    return ctx.invoke(self.callback, **ctx.params)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

Submitted by

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